

Your Success is our win Ex#2-6

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- ① Solve the following homogenous system of linear equations for non-trivial solutions, if possible.

$$\begin{aligned} x + 2y - 2z &= 0 \\ 2x + y + 5z &= 0 \\ 5x + 4y + 8z &= 0 \end{aligned}$$

sol

Where,

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 5 \\ 5 & 4 & 8 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$A \sim \begin{bmatrix} 1 & 2 & -2 \\ 2-2 & 1-4 & 5+4 \\ 5-5 & 4-10 & 8+10 \end{bmatrix} \begin{matrix} \\ R_2 - 2R_1 \\ R_3 - 5R_1 \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 9 \\ 0 & -6 & 18 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & 2 & -2 \\ 0 & (-3)(-1/3) & (9)(-1/3) \\ 0 & -6 & 18 \end{bmatrix} \begin{matrix} \\ (-1/3)R_2 \\ (-1/3)R_2 \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & -6 & 18 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & -6+6 & 18-18 \end{bmatrix} R_3 + 6R_2$$

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$$A \sim \begin{bmatrix} 0 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, we have

$$\begin{bmatrix} 0 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{--- (A)}$$

Here, Rank = 2, $n = 3$. Since Rank(A) = 2 < 3, So system has infinite solutions.

From (A) we get,

$$x + 2y - 2z = 0 \quad \text{--- (1)}$$

$$y - 3z = 0 \quad \text{--- (2)}$$

Take (2)

$$y - 3z = 0$$

$$y = 3z$$

Put y in (1)

$$x + 2(3z) - 2z = 0$$

$$x + 6z - 2z = 0$$

$$x = -4z$$

let,

$$z = k, k \neq 0$$

Now the solutions

$$\text{are } x = -4k, y = 3k$$

$$\text{and } z = k.$$

For $k=1$ we get,

$$x = -4, y = 3 \text{ \& } z = 1$$

For $k=2$ we get,

$$x = -8, y = 6 \text{ \& } z = 2$$

So non-trivial solutions

are $(-4, 3, 1)$ and $(-8, 6, 2)$ at $k=1$ & $k=2$ respectively

Q.P.

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$$(H) \quad x + 4y + 2z = 0$$

$$2x + y - 3z = 0$$

$$3x + 2y - 4z = 0$$

Sol

Where,

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 1 & -3 \\ 3 & 2 & -4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$A \sim \begin{bmatrix} 1 & 4 & 2 \\ 2-2 & 1-8 & -3-4 \\ 3-3 & 2-12 & -4-6 \end{bmatrix} \begin{matrix} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & 4 & 2 \\ 0 & -7 & -7 \\ 0 & -10 & -10 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & -10 & -10 \end{bmatrix} \begin{matrix} \\ (-1/7)R_2 \\ \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & -10+10 & -10+10 \end{bmatrix} \begin{matrix} \\ \\ R_3 + 10R_2 \end{matrix}$$

$$A \sim \begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus, we have

$$\begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{--- (A)}$$

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Here Rank = 2, $n = 3$. Since Rank (A) = 2 < 3, So the system has infinitely solutions.

From (A) we get,

$$x + 4y + 2z = 0 \quad \text{--- (1)}$$

$$y + z = 0 \quad \text{--- (2)}$$

Take (2)

$$y + z = 0$$

$$y = -z$$

Put y in (1)

$$x + 4(-z) + 2z = 0$$

$$x - 4z + 2z = 0$$

$$x = 2z$$

Let, $z = k$, $k \neq 0$

Now the solutions are

$$x = 2k, y = -k \text{ \& } z = k$$

For $k = 1$ we get,

$$x = 2, y = -1, z = 1$$

For $k = 2$ we get,

$$x = 4, y = -2, z = 2$$

So non-trivial solutions

are $(2, -1, 1)$ & $(4, -2, 2)$

at $k = 1$ & $k = 2$ respectively.

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Q2) Determine the consistency of non-homogeneous system of linear equations.

$$(i) \quad x - 2y - 2z = -1$$

$$2x + 3y + z = 1$$

$$5x - 4y - 3z = 1$$

Sol

We write the above system of equation of equation in matrix form.

$$AX = B$$

i.e.

$$\begin{bmatrix} 1 & -2 & -2 \\ 2 & 3 & 1 \\ 5 & -4 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Augmented matrix} = A_B = \begin{bmatrix} 1 & -2 & -2 & -1 \\ 2 & 3 & 1 & 1 \\ 5 & -4 & -3 & 1 \end{bmatrix}$$

Now, reduce it to echelon form.

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & -1 \\ 2-2 & 3+4 & 1+4 & 1+2 \\ 5-5 & -4+10 & -3+10 & 1+5 \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 5R_1 \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 7 & 5 & 3 \\ 0 & 6 & 7 & 6 \end{bmatrix}$$

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$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 1-6 & 5-7 & 3-6 \\ 0 & 6 & 7 & 6 \end{bmatrix} \quad R_2 \rightarrow R_2 + R_1$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 1 & -2 & -3 \\ 0 & 6 & 7 & 6 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 1 & -2 & -3 \\ 0 & 6-6 & 7+12 & 6+18 \end{bmatrix} \quad R_3 \rightarrow R_3 - 6R_2$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 19 & -24 \end{bmatrix}$$

Here,

$$\text{Rank of } A = 3 \text{ \& } \text{Rank of } A_B = 3$$

Thus,

$$\text{Rank of } A = \text{Rank of } A_B$$

Hence the system is consistent.

(ii) $x + 2x + z = 2$

$$2x + y + 2z = -1$$

$$2x + 3y - z = 9$$

∴

Write the above system of equation in matrix form.

$$AX = B$$

i.e.,

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$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 7 \end{bmatrix}$$

Here,

$$\text{Augmented matrix} = A_B = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & -1 \\ 2 & 3 & -1 & 9 \end{bmatrix}$$

Reduce it to echelon form.

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2-2 & 1-4 & 2-2 & -1-2 \\ 2-2 & 3-4 & -1-2 & 9-4 \end{bmatrix} \quad \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & -5 \\ 0 & -1 & -3 & 5 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 0 & 5/3 \\ 0 & -1 & -3 & 5 \end{bmatrix} \quad (-1/3) R_2$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 0 & 5/3 \\ 0 & 0 & -3 & 20/3 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

Here,

$$\text{Rank of } A = \text{Rank of } A_B$$

Hence, the system is consistent.

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- ① Solve the non-homogenous system of linear equations using matrix inversion method.

$$\begin{aligned} (1) \quad x + 2y + z &= 8 \\ 2x - y + z &= 3 \\ x + y - z &= 0 \end{aligned}$$

Sol

Write system of linear equations in form of matrix i.e. $AX = B$

Where,

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 8 \\ 3 \\ 0 \end{bmatrix}$$

Acc to matrix inversion method,

$$X = A^{-1}B$$

First we find A^{-1} .

$$\therefore A^{-1} = \frac{\text{adj of } A}{|A|}$$

$$|A| = \begin{vmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$|A| = 1(-1-1) - 2(-2-1) + 1(2+1)$$

$$|A| = 1(0) - 2(-3) + 1(3)$$

$$|A| = 6 + 3$$

$$|A| = 9$$

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$$\therefore \text{adj of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T$$

Co-factor of A:

$$\therefore A_{ij} = (-1)^{i+j} M_{ij}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= (-1)^2 (1-1)$$

$$= 0$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= (-1)^3 (-2-1)$$

$$= 3$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix}$$

$$= (-1)^4 (2+1)$$

$$= 3$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= (-1)^3 (-2-1)$$

$$= 3$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$$= (-1)^4 (-1-1)$$

$$= -2$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= (-1)^5 (1-2)$$

$$= 1$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix}$$

$$= (-1)^4 (2+1)$$

$$= 3$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}$$

$$= (-1)^5 (1-2)$$

$$= 1$$

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$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix}$$

$$= (-1)^4 (-1-4)$$

$$= -5$$

$$\text{adj of } A = \begin{bmatrix} 0 & 3 & 3 \\ 3 & -2 & 1 \\ 3 & 1 & -5 \end{bmatrix}^T$$

$$\text{adj of } A = \begin{bmatrix} 0 & 3 & 3 \\ 3 & -2 & 1 \\ 3 & 1 & -5 \end{bmatrix}$$

Now,

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} 0 & 3 & 3 \\ 3 & -2 & 1 \\ 3 & 1 & -5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/2 & -2/4 & 1/4 \\ 1/2 & 1/4 & -5/4 \end{bmatrix}$$

Now,

$$X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/2 & -2/4 & 1/4 \\ 1/2 & 1/4 & -5/4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0+1+0 \\ 1/2-2/2+0 \\ 1/2+1/2-0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Thus,

$$x=1, y=2 \text{ \& } z=3$$

is the required solution. $\square$

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$$\begin{aligned} \text{(ii)} \quad & 2x - y + 2z = 4 \\ & x + 10y - 3z = 10 \\ & -x + y + z = -6 \end{aligned}$$

SolWrite above system of linear equations in matrix form - i.e. $AX=B$

$$A = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 10 & -3 \\ -1 & 1 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 \\ 10 \\ -6 \end{bmatrix}$$

First we find A^{-1} .

$$A^{-1} = \frac{\text{adj of } A}{|A|}$$

$$|A| = \begin{vmatrix} 2 & -1 & 2 \\ 1 & 10 & -3 \\ -1 & 1 & 1 \end{vmatrix}$$

$$|A| = 2 \begin{vmatrix} 10 & -3 \\ 1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & -3 \\ -1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 1 & 10 \\ -1 & 1 \end{vmatrix}$$

$$|A| = 2(10+3) + 1(1-3) + 2(1+10)$$

$$|A| = 2(13) + 1(-2) + 2(11)$$

$$|A| = 26 - 2 + 22$$

$$|A| = 46$$

$$\text{adj of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T$$

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Co-factors of A:

$$\therefore A_{ij} = (-1)^{i+j} M_{ij}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 10 & -3 \\ 1 & 1 \end{vmatrix}$$

$$= (-1)^2 (10 + 3)$$

$$= 13$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 1 & -3 \\ -1 & 1 \end{vmatrix}$$

$$= (-1)^3 (1 - 3)$$

$$= 2$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 1 & 10 \\ -1 & 1 \end{vmatrix}$$

$$= (-1)^4 (1 + 10)$$

$$= 11$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} -1 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= (-1)^3 (-1 - 2)$$

$$= 3$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 2 \\ -1 & 1 \end{vmatrix}$$

$$= (-1)^4 (2 + 2)$$

$$= 4$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 2 & -1 \\ -1 & 1 \end{vmatrix}$$

$$= (-1)^5 (2 - 1)$$

$$= -1$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} -1 & 2 \\ 10 & -3 \end{vmatrix}$$

$$= (-1)^4 (3 - 20)$$

$$= -17$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 2 & 2 \\ 1 & -3 \end{vmatrix}$$

$$= (-1)^5 (-6 - 2)$$

$$= 8$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 2 & -1 \\ 1 & 10 \end{vmatrix}$$

$$= (-1)^6 (20 + 1)$$

$$= 21$$

$$\text{adj. of } A = \begin{bmatrix} 13 & 2 & 11 \\ 3 & 4 & -1 \\ -17 & 8 & 21 \end{bmatrix}^t$$

$$\text{adj. of } A = \begin{bmatrix} 13 & 2 & -17 \\ 3 & 4 & 8 \\ 11 & -1 & 21 \end{bmatrix}$$

Now,

$$A^{-1} = \frac{1}{46} \begin{bmatrix} 13 & 2 & -17 \\ 3 & 4 & 8 \\ 11 & -1 & 21 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 13/46 & 3/46 & -17/46 \\ 3/23 & 2/23 & 4/23 \\ 11/46 & -1/46 & 21/46 \end{bmatrix}$$

As we know that,

$$\therefore X = A^{-1} B$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 13/46 & 3/46 & -17/46 \\ 3/23 & 2/23 & 4/23 \\ 11/46 & -1/46 & 21/46 \end{bmatrix} \begin{bmatrix} 4 \\ 10 \\ -6 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (13/46)(4) + (3/46)(10) + (-17/46)(-6) \\ (3/23)(4) + (2/23)(10) + (4/23)(-6) \\ (11/46)(4) + (-1/46)(10) + (21/46)(-6) \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 26/23 + 15/23 + 51/23 \\ 4/23 + 20/23 - 24/23 \\ 22/23 - 15/23 - 63/23 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -2 \end{bmatrix}$$

Thus, $x = 4$, $y = 0$ and $z = -2$ is the required solution.

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Q. Solve the non-homogenous system of linear equation by Gauss elimination method.

$$\begin{aligned} (i) \quad & -x + y + z = 0 \\ & x + 2y = 5 \\ & -3x + 2y - z = -2 \end{aligned}$$

Sol

Write the above system of linear equations in matrix form i.e., $AX = B$

Where,

$$A = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 2 & 0 \\ -3 & 2 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 \\ 5 \\ -2 \end{bmatrix}$$

$$\text{Augmented matrix} = A_B = \begin{bmatrix} -1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 5 \\ -3 & 2 & -1 & -2 \end{bmatrix}$$

Reduce A_B into echelon form

$$AB \sim \begin{bmatrix} 1 & 2 & 0 & 5 \\ -1 & 1 & 1 & 0 \\ -3 & 2 & -1 & -2 \end{bmatrix} R_{12}$$

$$AB \sim \begin{bmatrix} 1 & 2 & 0 & 5 \\ -1+1 & 1+2 & 1+0 & 0+5 \\ -3+3 & 2+6 & -1+0 & -2+15 \end{bmatrix} \begin{matrix} \\ R_2 + R_1 \\ R_3 + 3R_1 \end{matrix}$$

$$AB \sim \begin{bmatrix} 1 & 2 & 0 & 5 \\ 0 & 3 & 1 & 5 \\ 0 & 8 & -1 & 13 \end{bmatrix}$$

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$$AB \sim \begin{bmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 1/3 & 5/3 \\ 0 & 8 & -1 & 13 \end{bmatrix} \begin{matrix} \\ (1/3)R_2 \\ \end{matrix}$$

$$AB \sim \begin{bmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 1/3 & 5/3 \\ 0 & 8-8 & -1-8(1/3) & 13-8(5/3) \end{bmatrix} R_3 - 8R_2$$

$$AB \sim \begin{bmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 1/3 & 5/3 \\ 0 & 0 & -11/3 & -1/3 \end{bmatrix}$$

$$AB \sim \begin{bmatrix} 1 & 2 & 0 & 5 \\ 0 & 1 & 1/3 & 5/3 \\ 0 & 0 & 1 & 1/11 \end{bmatrix}$$

From the above matrix we get equivalent matrix equation.

$$x + 2y + 0 = 5 \quad \text{--- (1)}$$

$$y + z/3 = 5/3 \quad \text{--- (2)}$$

$$z = 1/11 \quad \text{--- (3)}$$

Put z in (2)

$$\Rightarrow y + \frac{2}{3} = 5$$

$$\Rightarrow y + \frac{1}{22} = \frac{5}{3}$$

$$\Rightarrow y = \frac{5}{3} - \frac{1}{22}$$

$$\Rightarrow y = \frac{55-1}{33}$$

$$\Rightarrow y = \frac{54}{33}$$

$$\Rightarrow y = \frac{18}{11}$$

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Put y and z in ①

$$\Rightarrow x + 2y = 5$$

$$\Rightarrow x + 2\left(\frac{18}{11}\right) = 5$$

$$\Rightarrow x + \frac{36}{11} = 5$$

$$\Rightarrow x = 5 - \frac{36}{11}$$

$$\Rightarrow x = \frac{55 - 36}{11}$$

$$\Rightarrow x = \frac{19}{11}$$

Thus,

$x = \frac{19}{11}$, $y = \frac{18}{11}$ and $z = \frac{1}{11}$ are the required solutions.

Q2

$$(ii) \quad x + 2y + z = 8$$

$$2x - y + z = 3$$

$$x + y - z = 0$$

Sol.

Write the above system of linear equations in matrix form i.e., $AX = B$

where,

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 8 \\ 3 \\ 0 \end{bmatrix}$$

Now,

$$\text{Augmented Matrix} = A_B = \begin{bmatrix} 1 & 2 & 1 & 8 \\ 2 & -1 & 1 & 3 \\ 1 & 1 & -1 & 0 \end{bmatrix}$$

Reduced it into echelon form.

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 2 & -1 & 1 & 3 \\ 1 & 1 & -1 & 0 \end{bmatrix} \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

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$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & -5 & -1 & -13 \\ 0 & -1 & -2 & -8 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & -5 & -6(-1) & -13 - 6(-8) \\ 0 & -1 & -2 & -8 \end{bmatrix} \begin{array}{l} R_2 - 6R_3 \end{array}$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & 1 & 11 & 35 \\ 0 & -1 & -2 & -8 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & 1 & 11 & 35 \\ 0 & 1 & -2 & -8 \end{bmatrix} \begin{array}{l} R_3 + R_2 \end{array}$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & 1 & 11 & 35 \\ 0 & 0 & 9 & 27 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & 1 & 11 & 35 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{array}{l} (1/9) R_3 \end{array}$$

From above matrix we get another equivalent matrix equation.

$$x + 2y + z = 8 \quad \text{--- ①}$$

$$y + 11z = 35 \quad \text{--- ②}$$

$$z = 3 \quad \text{--- ③}$$

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Put $z = 1$ in (1)

$$\Rightarrow y + 11z = 35$$

$$\Rightarrow y + 11(1) = 35$$

$$\Rightarrow y = 35 - 11$$

$$\Rightarrow y = 24$$

Put $y = 2$ in (2)

$$\Rightarrow x + 2y + z = 8$$

$$\Rightarrow x + 2(2) + 1 = 8$$

$$\Rightarrow x = 8 - 5$$

$$\Rightarrow x = 3$$

Thus,

$x = 3, y = 2$ and $z = 1$ are the required solution.

Q10) Solve the non-homogeneous system of linear equations using Gauss-Jordan Method.

$$(1) \quad x - y + 4z = 4$$

$$2x + 2y - z = 2$$

$$3x - 2y + 3z = -7$$

Sol

Write above system of linear equation in matrix form i.e. $AX = B$

where,

$$A = \begin{bmatrix} 1 & -1 & 4 \\ 2 & 2 & -1 \\ 3 & -2 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 2 \\ -7 \end{bmatrix}$$

Now,

$$\text{Augmented Matrix} = A_B = \begin{bmatrix} 1 & -1 & 4 & 4 \\ 2 & 2 & -1 & 2 \\ 3 & -2 & 3 & -7 \end{bmatrix}$$

Reduce it into echelon form

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$$A_B \sim \begin{bmatrix} 1 & -1 & 4 & 4 \\ 2 & 2 & -1 & 2 \\ 3 & -2 & 3 & -7 \end{bmatrix} \begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$A_B \sim \begin{bmatrix} 1 & -1 & 4 & 4 \\ 0 & 4 & -9 & -6 \\ 0 & 1 & -9 & -15 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -1 & 4 & 4 \\ 0 & 4 & -9 & -6 \\ 0 & 1 & -9 & -15 \end{bmatrix} \begin{array}{l} R_2 - 4R_3 \end{array}$$

$$A_B \sim \begin{bmatrix} 1 & -1 & 4 & 4 \\ 0 & 1 & 18 & 39 \\ 0 & 1 & -9 & -15 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -1 & 4 & 4 \\ 0 & 1 & 18 & 39 \\ 0 & 0 & -27 & -54 \end{bmatrix} \begin{array}{l} R_2 - R_3 \end{array}$$

$$A_B \sim \begin{bmatrix} 1 & -1 & 4 & 4 \\ 0 & 1 & 18 & 39 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{array}{l} (\frac{1}{27})R_3 \end{array}$$

$$A_B \sim \begin{bmatrix} 1 & -1 & 4 & 4 \\ 0 & 1 & 18 & 39 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{array}{l} R_1 + R_2 \end{array}$$

$$A_B \sim \begin{bmatrix} 1 & 0 & 22 & 43 \\ 0 & 1 & 18 & 39 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

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$$A_B \sim \begin{bmatrix} 1 & 0 & 22-22 & 43-24 \\ 0 & 1 & 18-18 & 39-38 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{matrix} R_1 - 2R_3 \\ R_2 - 18R_3 \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

From above matrix we get equivalent matrix equation.

$$0x + 0y + 0z = -1$$

$$0x + y + 0z = 3$$

$$0x + 0y + z = 2$$

Thus,

$x=1, y=3$ and $z=2$ is the required

Solution. $\frac{11}{2}$

(ii) $2x + 2y - z = 4$

$x - 2y + z = 2$

(6) $x + y = 0$

Sol

Write above system of linear equations in matrix form i.e. $AX=B$

Where,

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & -2 & 1 \\ 1 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

94

Augmented Matrix = $A_B = \begin{bmatrix} 2 & 2 & -1 & 4 \\ 1 & -2 & 1 & 2 \\ 1 & 1 & 0 & 0 \end{bmatrix}$

Reduce it into echelon form

$$A_B \sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 2 & 2 & -1 & 4 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{matrix} R_{12} \\ \\ \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 2-2 & 2+4 & -1-2 & 4-4 \\ 1-1 & 1+2 & 0-1 & 0-2 \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - R_1 \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 6 & -3 & 0 \\ 0 & 3 & -1 & -2 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -1/2 & 0 \\ 0 & 3 & -1 & -2 \end{bmatrix} \begin{matrix} \\ (1/6)R_2 \\ \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -1/2 & 0 \\ 0 & 3-3 & -1-3(-1/2) & -2-0 \end{bmatrix} \begin{matrix} \\ \\ R_3 - 3R_2 \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 1/2 & -2 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 1 & -4 \end{bmatrix} \begin{matrix} \\ \\ (2)R_3 \end{matrix}$$

(95)

$$A_B \sim \begin{bmatrix} 1 & -2+2 & 1+2(-1/2) & 2+0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 1 & -4 \end{bmatrix} R_1 + 2R_2$$

$$A_B \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 1 & -4 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -1/2+1/2 & 0+2 \\ 0 & 0 & 1 & -4 \end{bmatrix} R_2 + (1/2)R_3$$

$$A_B \sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -4 \end{bmatrix}$$

From above matrix we get equivalent matrix equation.

$$x + 0y + 0z = 2$$

$$0x + y + 0z = -2$$

$$0x + 0y + z = -4$$

Thus,

$x=2, y=-2$ and $z=-4$ is the required solution.

ES

(96)

Q#9) Solve the non-homogenous system of linear equations using Cramer's Rule.

$$(i) \quad x - 2y + z = 2$$

$$2x + 2y - z = 4$$

$$x + y + 0 \cdot z = 0$$

Sol

Write the above system of linear equations in matrix form i.e. $AX = B$

$$\text{i.e., } \begin{bmatrix} 1 & -2 & 1 \\ 2 & 2 & -1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}$$

Here,

$$a_1 = 1, b_1 = -2, c_1 = 1, d_1 = 2$$

$$a_2 = 2, b_2 = 2, c_2 = -1, d_2 = 4$$

$$a_3 = 1, b_3 = 1, c_3 = 0, d_3 = 0$$

For $|A|$:

$$|A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 2 & -1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$|A| = 1 \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} + 1 \begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix}$$

$$|A| = 1(0+1) + 2(0+1) + 1(2-2)$$

$$|A| = 3$$

For $|A_x|$:

$$|A_x| = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$|A_x| = \begin{vmatrix} 2 & -2 & 1 \\ 4 & 2 & -1 \\ 0 & 1 & 0 \end{vmatrix}$$

$$|A_x| = 2 \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} - 2 \begin{vmatrix} 4 & -1 \\ 0 & 0 \end{vmatrix} + 1 \begin{vmatrix} 4 & 2 \\ 0 & 1 \end{vmatrix}$$

$$|A_x| = 2(0+1) + 2(0) + 1(4-0)$$

$$|A_x| = 6$$

$$|A_y| = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$|A_y| = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 4 & -1 \\ 1 & 0 & 0 \end{vmatrix}$$

$$|A_y| = 1(4 \cdot 0 - 1 \cdot 2) - 2(1 \cdot 0 - 1 \cdot 1) + 1(2 \cdot 0 - 1 \cdot 1)$$

$$|A_y| = 1(0) - 2(0+1) + 1(0-1)$$

$$|A_y| = -6$$

$$|A_z| = \begin{vmatrix} 1 & -2 & 2 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$|A_z| = \begin{vmatrix} 1 & -2 & 2 \\ 2 & 2 & 4 \\ 1 & 1 & 0 \end{vmatrix}$$

$$|A_z| = 1(2 \cdot 4 - 2 \cdot 2) + 2(2 \cdot 0 - 2 \cdot 1) + 2(2 \cdot 1 - 1 \cdot 1)$$

$$|A_z| = 1(-4) + 2(-2) + 2(2-1)$$

$$|A_z| = -12$$

By applying Cramer's rule we get,

$$\Rightarrow x = \frac{|A_x|}{|A|} \quad \Rightarrow y = \frac{|A_y|}{|A|} \quad \Rightarrow z = \frac{|A_z|}{|A|}$$

$$\Rightarrow x = \frac{6}{3} \quad \Rightarrow y = \frac{-6}{3} \quad \Rightarrow z = \frac{-12}{3}$$

$$\Rightarrow x = 2 \quad \Rightarrow y = -2 \quad \Rightarrow z = -4$$

(16)

Thus,

$x = 2$, $y = -2$ and $z = -4$ is the required solution.

Q.8

(17)

$$(17) \quad x - 2y + 0 \cdot z = -4$$

$$3x + y + 0 \cdot z = -5$$

$$2x + 0 \cdot y + z = -1$$

Sol

Write above system of linear equations in matrix form i.e. $AX = B$

Where,

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} -4 \\ -5 \\ -1 \end{bmatrix}$$

Here,

$$a_1 = 1, b_1 = -2, c_1 = 0, d_1 = -4$$

$$a_2 = 3, b_2 = 1, c_2 = 0, d_2 = -5$$

$$a_3 = 2, b_3 = 0, c_3 = 1, d_3 = -1$$

For $|A|$:

$$|A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & -2 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{vmatrix}$$

$$|A| = 1(1 \cdot 0 - 2 \cdot 0) + 2(3 \cdot 0 - 0 \cdot 2) + 0(2 \cdot 0 - 1 \cdot 1)$$

$$|A| = 1(0) + 2(0-0) + 0(0-2)$$

$$|A| = 7$$

For $|A_x|$:

$$|A_x| = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$|A_x| = \begin{vmatrix} -4 & -2 & 0 \\ -5 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$$

$$|A_x| = -4(1 \cdot 0 - 2 \cdot 0) + 2(-5 \cdot 0 - 0 \cdot 1) + 0(-1 \cdot 0 - 1 \cdot 1)$$

$$|A_x| = -4(0) + 2(0-0) + 0(0-1)$$

$$|A_x| = -14$$

(18)

(Q9)

For $|A_y|$:

$$|A_y| = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$|A_y| = \begin{vmatrix} 1 & -4 & 0 \\ 3 & -5 & 0 \\ 2 & -1 & 1 \end{vmatrix}$$

$$|A_y| = 1 \begin{vmatrix} -5 & 0 \\ -1 & 1 \end{vmatrix} + 4 \begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} + 0 \begin{vmatrix} 3 & -5 \\ 2 & -1 \end{vmatrix}$$

$$|A_y| = 1(-5+0) + 4(3-0) + 0(-3+10)$$

$$|A_y| = -5 + 12$$

$$|A_y| = 7$$

For $|A_z|$:

$$|A_z| = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

$$|A_z| = \begin{vmatrix} 1 & -2 & -4 \\ 3 & 1 & -5 \\ 2 & 0 & -1 \end{vmatrix}$$

$$|A_z| = 1 \begin{vmatrix} -5 & -4 \\ 2 & -1 \end{vmatrix} + 2 \begin{vmatrix} 3 & -5 \\ 2 & -1 \end{vmatrix} - 4 \begin{vmatrix} 3 & 1 \\ 2 & 0 \end{vmatrix}$$

$$|A_z| = 1(-5-0) + 2(-3+10) - 4(0-2)$$

$$|A_z| = -1 + 14 + 8$$

$$|A_z| = 21$$

By applying cramer's rule we get:

$$\Rightarrow x = \frac{|A_x|}{|A|}$$

$$\Rightarrow x = \frac{-14}{7}$$

$$\Rightarrow x = -2$$

$$\Rightarrow y = \frac{|A_y|}{|A|}$$

$$\Rightarrow y = \frac{7}{7}$$

$$\Rightarrow y = 1$$

$$\Rightarrow z = \frac{|A_z|}{|A|}$$

$$\Rightarrow z = \frac{21}{7}$$

$$\Rightarrow z = 3$$

Thus,

$x = -2$, $y = 1$ and $z = 3$ is the required solution